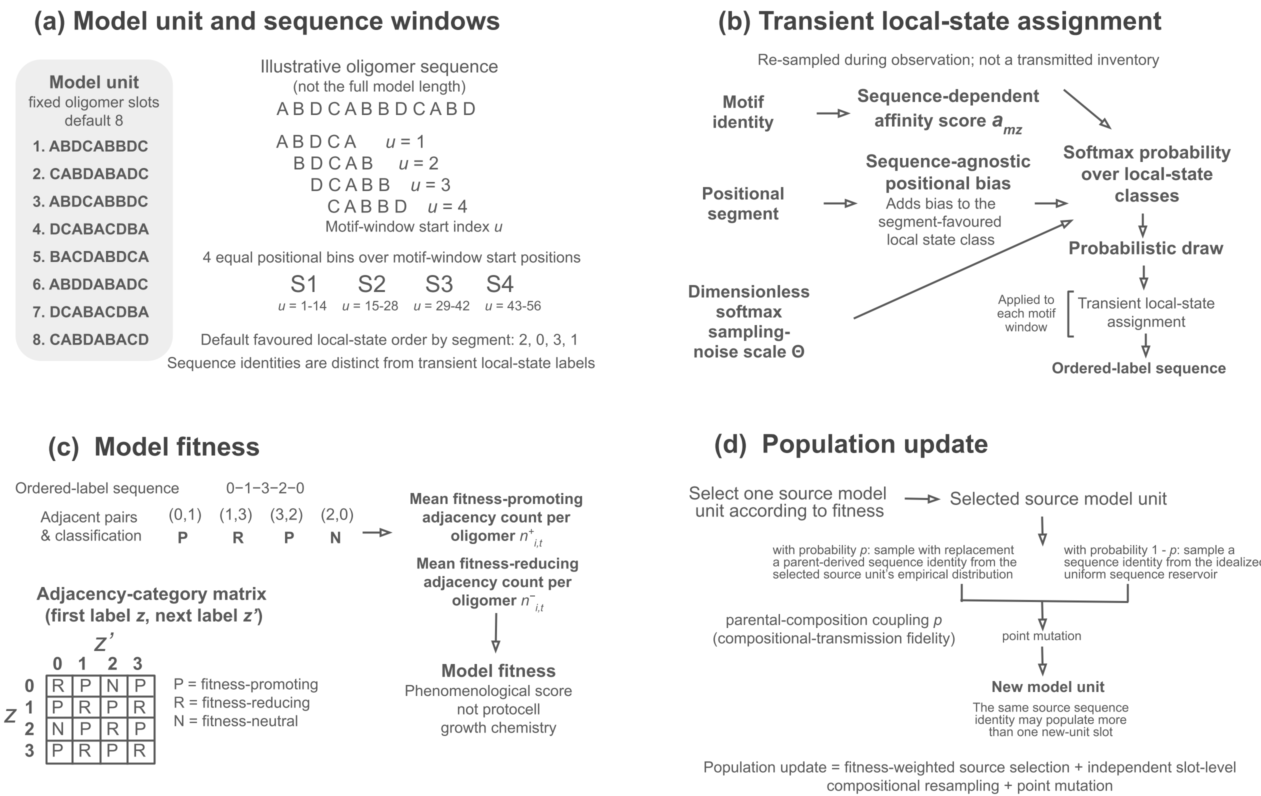

(a) Model unit and sequence windows
Model unit
fixed oligomer slots
default 8
1. ABDCABBDC
2. CABDABADC
3. ABDCABBDC
4. DCABACDBA
5. BACDABDCA
6. ABDDABADC
7. DCABACDBA
8. CABDABACD
Illustrative oligomer sequence
(not the full model length)
A B D C A B B D C A B D
A B D C A u = 1
B D C A B u = 2
D C A B B u = 3
C A B B D u = 4
Motif-window start index u
4 equal positional bins over motif-window start positions
S1 S2 S3 S4
u = 1-14 u = 15-28 u = 29-42 u = 43-56
Default favoured local-state order by segment: 2, 0, 3, 1
Sequence identities are distinct from transient local-state labels
(b) Transient local-state assignment
Re-sampled during observation; not a transmitted inventory
Motif identity
Sequence-dependent affinity score a_{mz}
Positional segment
Sequence-agnostic positional bias
Adds bias to the segment-favoured local state class
Dimensionless softmax sampling-noise scale Θ
Softmax probability over local-state classes
Probabilistic draw
Applied to each motif window
Transient local-state assignment
Ordered-label sequence
(c) Model fitness
Ordered-label sequence 0−1−3−2−0
Adjacent pairs & classification
(0,1) (1,3) (3,2) (2,0)
P R P N
Mean fitness-promoting adjacency count per oligomer $n^{+}_{i,t}$
Mean fitness-reducing adjacency count per oligomer $n^{-}_{i,t}$
Model fitness
Phenomenological score not protocell growth chemistry
Adjacency-category matrix (first label z, next label z′)
z′
0 1 2 3
z
0 R P N P
1 P R P R
2 N P R P
3 P R P R
P = fitness-promoting
R = fitness-reducing
N = fitness-neutral
(d) Population update
Select one source model unit according to fitness
Selected source model unit
with probability p: sample with replacement a parent-derived sequence identity from the selected source unit's empirical distribution
with probability 1 - p: sample a sequence identity from the idealized uniform sequence reservoir
parental-composition coupling p
(compositional-transmission fidelity)
point mutation
New model unit
The same source sequence identity may populate more than one new-unit slot
Population update = fitness-weighted source selection + independent slot-level compositional resampling + point mutation